CHEMISTRY ADAPTLY

PROPERTIES OF SMALL MOLECULES

KEY VOCABULARY

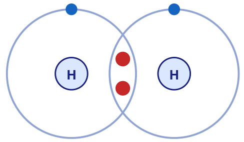
Simple molecule: a small group of atoms covalently bonded, e.g. CO_2.
Intermolecular force: weak attraction between separate molecules.
Volatile: evaporates easily — low boiling point.
Insulator: does not conduct electricity.

COMMON MISCONCEPTIONS

~~Bigger molecules break their bonds more easily.~~
✓ Bigger molecules have stronger intermolecular forces.
~~Melting breaks covalent bonds.~~
✓ Melting only overcomes weak intermolecular forces.
~~Molecular substances are good conductors.~~
✓ They have no free charges, so they don't conduct.

COVALENT MOLECULE

A SHARED PAIR OF ELECTRONS



Each H shares one electron → both have a full shell (2)

Strong bonds WITHIN molecules; weak forces BETWEEN them.

1 STATE AT RTP

State why many simple molecules are **gases or liquids** at room temperature. **[1 mark]**

2 WHAT IS OVERCOME

State what is overcome when a simple molecular substance boils. **[1 mark]**

3 TREND IN BP

Explain why boiling point increases as molecules get larger. **[2 marks]**

4 CONDUCTIVITY

Explain why simple molecular substances do not conduct electricity. **[2 marks]**

5 FILL THE GAPS

Complete the sentence using the word bank. **[2 marks]**

Simple molecules have _____ boiling points because the _____ forces are weak.

intermolecular | covalent | low | high

6 COMPARE

Suggest why water has a higher boiling point than methane. **[2 marks]**

***TIP:** Boiling point depends on the strength of forces BETWEEN molecules.*

✓ Think: Why doesn't the size of the covalent bond change the boiling point?